POWER BI

Power BI Interview Notes

Data model, the 10 DAX functions you will actually use, and the questions that come up most.

Model it as a star schema

One fact table (sales, orders) in the middle. Dimension tables (date, product, customer) around it. One-to-many relationships, single direction filters. Always add a proper Date table and mark it as a date table.

Measure vs calculated column

	Calculated column	Measure
Computed	At refresh, row by row	At query time
Stored	Yes, uses memory	No
Context	Row context	Filter context
Use for	Slicing, grouping	Totals, ratios, KPIs

The 10 DAX functions

```
Total Sales   = SUM(Sales[Amount])
Sales North   = CALCULATE([Total Sales], Product[Region] = "North")
% of Total    = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Product)))
Revenue       = SUMX(Sales, Sales[Qty] * Sales[Price])
Sales LY      = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
YoY %         = DIVIDE([Total Sales] - [Sales LY], [Sales LY])
Sales YTD     = TOTALYTD([Total Sales], 'Date'[Date])
Last Month    = CALCULATE([Total Sales], DATEADD('Date'[Date], -1, MONTH))
Big Orders    = CALCULATE([Total Sales], FILTER(Sales, Sales[Amount] > 10000))
Product Rank  = RANKX(ALL(Product[Name]), [Total Sales])
Category      = RELATED(Product[Category])   -- in a calculated column
```

Power Query steps you should know

- Remove duplicates, change type, trim, replace values.
- Unpivot columns (wide months to long format).
- Merge queries (join) and Append queries (union).
- Group By and conditional columns.

The 6 visuals that cover most dashboards

KPI card, line chart (trend), bar chart (comparison), matrix (detail), slicer (filter), map only when geography matters.

15 most asked questions

1. Filter context vs row context.
2. What does CALCULATE do?
3. ALL vs ALLEXCEPT vs REMOVEFILTERS.
4. Import vs DirectQuery vs Live connection.
5. What is Row Level Security and how do you set it up?
6. Why do you need a Date table?
7. Star vs snowflake schema.
8. Many-to-many relationships and their risk.
9. Bi-directional filter: when and why not.
10. SUM vs SUMX.
11. How to improve a slow report.
12. Drill down vs drill through.
13. Bookmarks and what they are used for.
14. Scheduled refresh and the gateway.
15. Walk me through a dashboard you built.

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